Reporting Mandates, Stock Liquidity, and the Mix of Regulation

Abstract

We study how reporting mandates relate to stock liquidity across financial and nonfinancial disclosure regimes. Our setting combines a global market panel with original reports that establish coverage and reporting dates for nine regulation families. We distinguish the average family from the regulatory exposure of large issuers. First-year log-spread comparisons are -0.078 under equal family weights and +0.190 under market-capitalisation weights. Audit reporting accounts for 87.6% of the latter weight; a family-by-family decomposition explains the difference between the aggregates. Seven families support longer-window comparisons. Earlier disclosure, different liquidity measures and pre-policy trends limit a common causal interpretation. We also examine international adoption using lagged competition, learning, emulation and external-dependence measures. No coefficient survives the full multiplicity correction. The evidence highlights policy composition and the information already available to investors when assessing reporting regulation.

1. Introduction

Investors need information to assess the prospects and risks of the firms they trade. Reporting mandates can make that information more widely available and easier to compare. Governments have extended these requirements well beyond financial statements to environmental performance, governance, supply chains and payments to governments. For accounting research, this creates a natural question: do the liquidity benefits associated with one reporting regime extend to a broader set of mandates?

The answer is not clear in advance. A requirement can reduce information differences between traders, but its contribution depends on what firms already report, who receives the disclosure and which firms must respond. A confidential tax return supplies information to the authorities without necessarily informing investors. A new governance code can require a response even where the underlying practice is already common. Mandatory reporting therefore need not mark the first arrival of information in the market.

We follow Krueger et al. (2023) in linking reporting rules to stock liquidity and then examining why the aggregate differs across settings. Liquidity matters to investors through trading costs and provides a common outcome for financial and nonfinancial disclosure. Our extension treats the mix of regulation as part of the research question. Equal family weights describe the average supported reporting regime; market-capitalisation weights give greater influence to regimes covering larger sampled issuers. Neither weighting scheme is selected for the direction of its result.

The main finding concerns this distinction. The two first-year spread comparisons differ by 0.268 log points while holding the family estimates fixed. The market-capitalisation aggregate is dominated by auditor reporting, with an effective family count of 1.30 rather than nine. This explains why an average across types of regulation can give a different answer from an average weighted toward large issuers. We complement the capital-market analysis with the diffusion framework of Koning, Mertens and Roosenboom (2018).

2. Sample and research design

The market panel contains 1,055,142 security-years in 90 jurisdictions from 2000 to 2025. We measure liquidity using the log relative quoted spread, log Amihud price impact and the fraction of observed returns equal to zero. Each proxy requires 120 valid observations; logged proxies are winsorised annually at the first and ninety-ninth percentiles. Higher values indicate lower liquidity. Historical identifiers link the selected reporting parents to ordinary shares; vendor quote identifiers do not by themselves establish legal coverage.

Table 1 describes nine families for which issuer reports or filing evidence support a comparison. Full reviews of previously unresolved SEC filings separate parent and subsidiary facts. SEC indices and original XBRL instances establish format dates; original bank reports and dated issuer records establish reporting years. The UK carbon study uses four original FY2013 annual reports recovered from the all-data archive. The 2013 law requires quoted-company greenhouse-gas disclosure; this is distinct from comprehensive E/S/G reporting and later climate-risk rules. Prior annual reports document information already available. Approval dates and archive stamps are never substituted for publication.

Eligibility and observed reporting remain distinct; seven CAM filings lack confirmed report text and enter a separate sensitivity. For each family and reporting cohort, we compare the change in treated issuers with the change in comparison issuers relative to year g-1. Calendar-based comparisons use known later implementers, with thirty issuers per country and three comparison markets. Report-based comparisons use documented reporting obligations and period-specific comparison evidence, requiring three treated and twenty comparison issuers. The first-year design uses g-1 and g. The longer-window design fixes support from g-3 through g+2 and averages years 0-2; it covers seven families.

Within each study, prior-year GDP is allocated across the fixed country issuer universe, and cohort estimates are combined using treated mass. Across families, we report both equal weights and prior-year treated-issuer market capitalisation. All three outcomes use the same supported families; banking enters once. Source-selected US comparisons do not certify every foreign subsidiary or venue exemption. Repeated issuer-policy exposures are not a unique-firm investment portfolio. Sparse common policy waves, cross-country and industry differences, and earlier disclosure constrain causal inference.

3. Liquidity and policy composition

Table 2 reports both aggregates. Across nine report-based families, first-year spreads are -0.078 under equal weights and +0.190 under market-capitalisation weights. Figure 1 shows the underlying comparisons and economic weights. Audit reporting receives 87.6% of market-capitalisation weight, compared with 11.1% under equal weights. The decomposition attributes the entire aggregate difference to these observed weight changes; it does not change any family coefficient.

Among the seven longer-window families, spread comparisons are -0.054 and -0.066. Price-impact comparisons are +0.071 and +0.159. Figure 2 shows the pre-policy differences as well as subsequent outcomes. These measures do not support a uniform improvement in trading conditions. The narrower longer-window sample also prevents interpreting the two horizons as one event path. The broader geographic panel in Table 2 is a separate exposure comparison, because country assignment alone does not establish issuer eligibility.

4. International diffusion

Following Koning et al., we ask whether adoption is associated with earlier implementation among trade peers, prior outcomes in neighbouring countries, cultural and institutional peers, or external financial dependence. All signals precede the focal decision. The corrected UK carbon date is incorporated into the adoption-risk panel and the lagged peer measures. Across 1,330 timing and status designs, 433 focal coefficients have stable estimates and 0 survive the full Holm correction. These exploratory associations do not establish a diffusion mechanism. Taken together, the analyses show why the covered population, policy mix and prior information must accompany any claim about the capital-market consequences of reporting mandates.

Table 1. Implemented reporting regimes and issuer comparisons

Family (study cohort)	First-year Treated / comparison	Balanced Treated / comparison	Log spread	Log price impact	Zero-return fraction
Auditor CAM (2019-21)	1,863 / 2,988	Unsupported	+0.230	+0.240	+0.0065
Early XBRL (2009)	6 / 62	Unsupported	-0.329	-0.417	-0.0004
Conflict minerals (2014)	3 / 70	3 / 45	-0.085	+0.249	+0.0071
Bank full reports (2015)	5 / 21	5 / 20	+0.066	+0.051	+0.0156
Canadian IFRS (2012)	4 / 42	4 / 39	-0.235	+0.132	-0.0096
German E/S/G (2018)	3 / 35	3 / 31	-0.217	-0.235	-0.0124
Extractive payments (2016)	7 / 35	7 / 24	-0.008	+0.128	+0.0044
UK Code revision (2012)	5 / 40	5 / 37	-0.155	-0.035	+0.0003
UK greenhouse gases (2014)	4 / 43	4 / 33	+0.029	-0.035	-0.0056
Alternative bank phase (2014)	23 / 21	22 / 20	+0.035	-0.343	+0.0006

First-year estimates use GDP weights within each study. Counts are issuer-cohort observations; comparisons can recur across CAM cohorts. The first nine rows enter the principal aggregate. The final row replaces the bank-full-report study in a separate sensitivity; banking is never counted twice. UK governance is a 2010 Code revision, not initial governance reporting. UK carbon is mandatory greenhouse-gas disclosure, not comprehensive E/S/G or later TCFD reporting. These source-selected studies do not cover every eligible issuer or all seventeen attempted families. CAM timing follows documented eligibility. Seven source-admitted filings lack confirmed CAM text; a stricter content comparison appears in the supplement.

Coverage and information already available

First statutory reporting is distinguished from first-ever disclosure. The UK carbon reports contain earlier-period comparatives; Drax also faced emissions-trading requirements. The governance study concerns a Code revision. Extractive companies had earlier voluntary tax disclosures. Fresnillo company 06344120 files its statutory report on 16 June 2016; all four official payloads agree. Its public report was available on 15 June.

Dates and source support

The four carbon issuers are BAE, Pearson, GKN and Drax. Pearson publishes on 24 March 2014, GKN on 27 March and Drax on 19 March; BAE retains March 2014 precision. Original reports establish the historical parent and listing, reporting period and disclosures. The comparison review examines 343 full SEC submissions from the all-data archive, with direct foreign-regime flags withheld. Positive source evidence does not certify all historical group and venue exemptions.

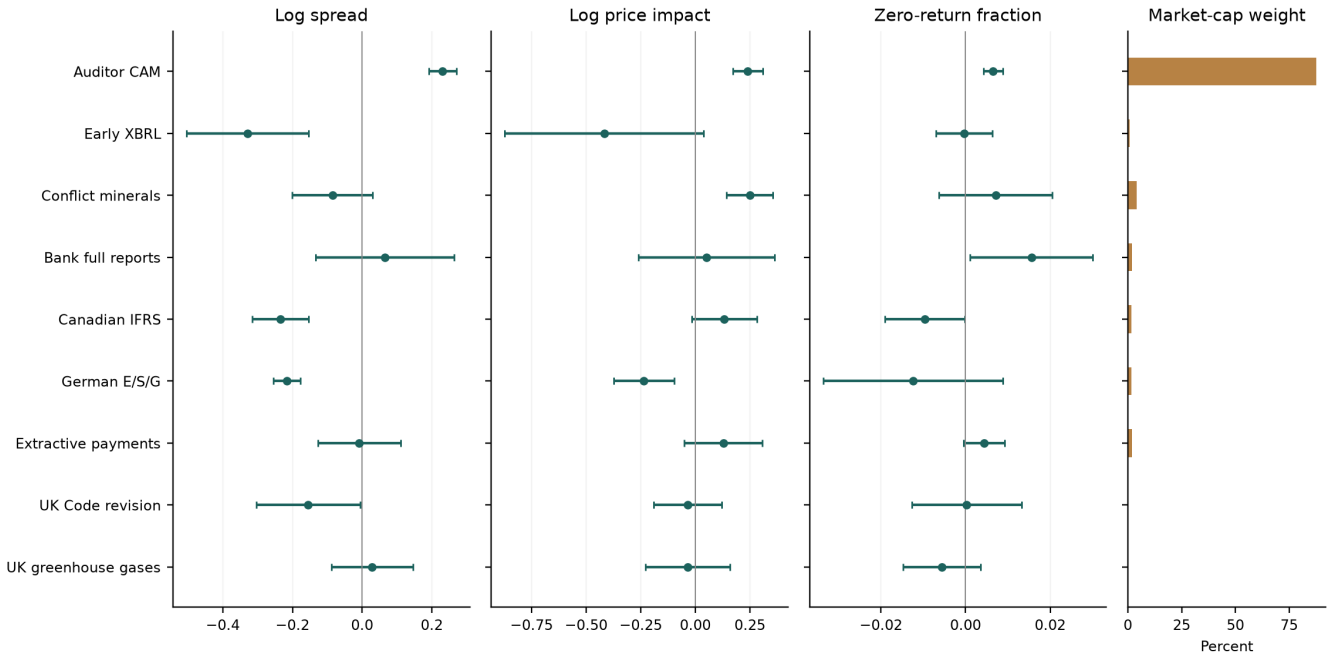
Table 2. Liquidity comparisons under both family weights

Population, horizon and liquidity measure	Equal family Estimate [interval]	Market-cap family Estimate [interval]
A. Geographic exposure First year (15 families): Log spread	-0.029 [-0.068, +0.011]	+0.021 [-0.021, +0.063]
A. Geographic exposure First year (15 families): Log price impact	-0.045 [-0.087, -0.003]	+0.019 [-0.050, +0.087]
A. Geographic exposure First year (15 families): Zero-return fraction	-0.0082 [-0.0112, -0.0052]	-0.0027 [-0.0065, +0.0012]
A. Geographic exposure Years 0-2 (9 families): Log spread	-0.017 [-0.078, +0.044]	+0.045 [-0.035, +0.125]
A. Geographic exposure Years 0-2 (9 families): Log price impact	-0.013 [-0.081, +0.054]	+0.096 [-0.024, +0.216]
A. Geographic exposure Years 0-2 (9 families): Zero-return fraction	-0.0006 [-0.0064, +0.0053]	+0.0070 [-0.0070, +0.0210]
B. Report-based issuer studies First year (9 families): Log spread	-0.078 [-0.123, -0.033]	+0.190 [+0.154, +0.225]
B. Report-based issuer studies First year (9 families): Log price impact	+0.009 [-0.062, +0.079]	+0.219 [+0.159, +0.280]
B. Report-based issuer studies First year (9 families): Zero-return fraction	+0.0007 [-0.0035, +0.0049]	+0.0060 [+0.0039, +0.0081]
B. Report-based issuer studies Years 0-2 (7 families): Log spread	-0.054 [-0.124, +0.016]	-0.066 [-0.144, +0.012]
B. Report-based issuer studies Years 0-2 (7 families): Log price impact	+0.071 [-0.026, +0.167]	+0.159 [+0.062, +0.255]
B. Report-based issuer studies Years 0-2 (7 families): Zero-return fraction	-0.0019 [-0.0065, +0.0027]	+0.0010 [-0.0033, +0.0053]

GDP weights within families. Higher outcomes indicate lower liquidity. Panel A describes individual outcomes under geographic policy exposure, not verified legal coverage; intervals condition on country sampling. Panel B intervals describe issuer sampling, retaining shared-issuer covariance; they are not causal policy confidence intervals. Small common policy waves, comparison selection, spillovers and pre-policy differences remain material. Family support differs across horizons. All outcomes and both weights are retained. The supplement reports every family estimate, alternative within-family weights, banking phases, support failure and single-family omission; unsupported families are not assigned zero.

Figure 1. Reporting families, liquidity and economic weights

First mandatory-reporting-year comparisons, GDP weighted within studies. All nine supported economic families and all three outcomes appear. Bars describe conditional issuer sampling, not causal policy uncertainty. The right panel gives each family its share of sampled treated-issuer capitalisation measured before reporting. Equal family weights are 1/9. Small treated groups and different market and industry exposures remain relevant.

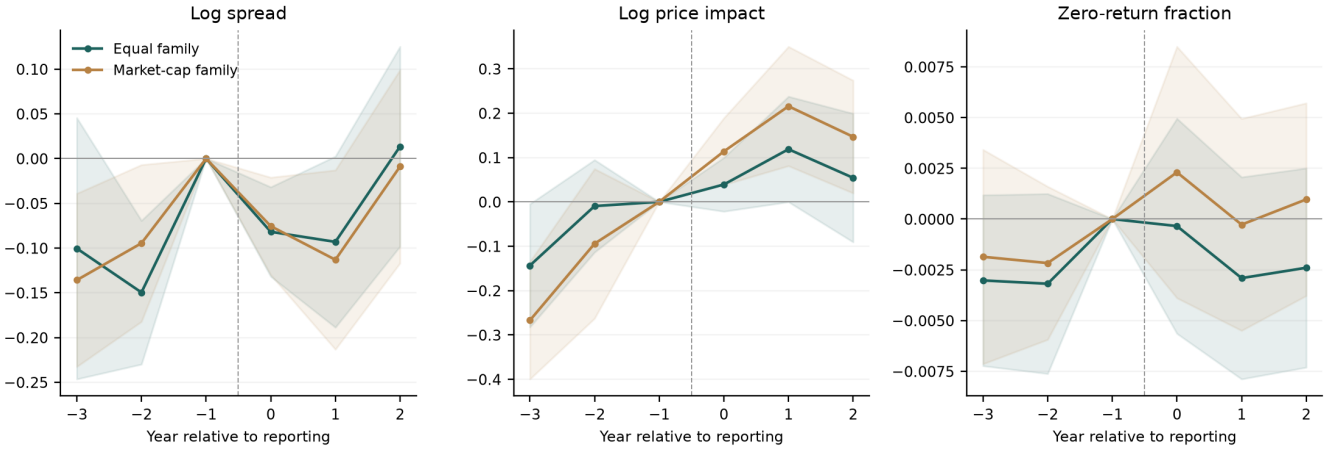


What the two weights represent

Equal family weights give each supported reporting regime the same influence. Capitalisation weights instead describe the mix of regulation associated with the sampled treated issuers' economic size. Audit reporting receives 87.6% of that weight. The exact change from the equal-weighted estimate is +0.268 log points. Neither aggregate can be interpreted as the effect of every reporting mandate on every listed firm. All within-family alternatives and all single-family omissions remain in the supplement.

Figure 2. Liquidity before and after mandatory reporting

Seven report-based families with fixed g-3 through g+2 support: banks, conflict minerals, IFRS, German E/S/G, extractive payments, UK revised-Code reporting and UK greenhouse gases. Year -1 is the reference. Shading describes pointwise issuer sampling, retaining shared-issuer covariance. Pre-policy differences are visible and constrain a causal interpretation. These profiles cannot be joined to the nine-family first-year aggregate as a single event path.



Diffusion design and interpretation

Competition uses lagged implementation among export-pattern peers. Learning uses prior outcomes among geographic neighbours; emulation uses language, legal-origin and IGO peers. External dependence uses IMF credit and aid. Timing and status regressions include family-year strata, country-clustered covariance and, where specified, lagged income, growth and trade controls. Unknown policy histories remain unknown. The 2013 UK carbon correction is carried through the risk sets and every lagged signal. A revised governance code or issuer filing date does not create another initial national governance adoption.

References and primary sources

Krueger, P., Z. Sautner, D. Y. Tang and R. Zhong. 2023. The Effects of Mandatory ESG Disclosure Around the World. Supplied working paper; Journal of Accounting Research (2024), doi:10.1111/1475-679X.12548.

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UK SI 2013/1970, regulations 1(4) and 7(3)(e); Schedule 7, Part 7. Original FY2013 BAE, Pearson, GKN and Drax annual reports; FY2012 prior reports; issuer publication announcements and archives.

Companies House Extractives Service: Fresnillo plc, company 06344120, FY2015 package filed 16 June 2016; Rio Tinto plc 00719885, 29 June; Anglo American plc 03564138, 30 June; Ferrexpo plc 05432915, 28 June. Four payload CSVs per parent preserve period, identity and filing date.

Primary reporting standards, original issuer reports and full legal chronology, source hashes, comparison evidence and complete results: https://hunternbh.github.io/policy-diffusion/